

Mathematics as verbal behavior[☆]

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ABSTRACT

“Behavior which is effective only through the mediation of other persons has so many distinguishing dynamic and topographical properties that a special treatment is justified and indeed demanded” (Skinner, 1957, p. 2). Skinner’s demand for a special treatment of verbal behavior can be extended *within* that field to domains such as music, poetry, drama, and the topic of this paper: mathematics. For centuries, mathematics has been of special concern to philosophers who have continually argued to the present day about what some deem its “special nature.” Two interrelated principal questions have been: (1) Are the subjects of mathematical interest pre-existing in some transcendental realm and thus are “discovered” as one might discover a new planet; and (2) Why is mathematics so effective in the practices of science and engineering even though originally such mathematics was “pure” with applications neither contemplated or even desired? I argue that considering the actual practice of mathematics in its history and in the context of acquired verbal behavior one can address at least some of its apparent mysteries. To this end, I discuss some of the structural and functional features of mathematics including verbal operants, rule- and contingency-modulated behavior, relational frames, the shaping of abstraction, and the development of intuition. How is it possible to understand Nature by properly talking about it? Essentially, it is because nature taught us how to talk.

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1. Introduction

“It is still an unending source of surprise for me to see how a few scribbles on a blackboard or on a sheet of paper could change the course of human affairs”. Stanislaw Ulam.

In this paper I am revisiting topics I have wrestled with off and on over decades, but an opportunity to speak at the Society for the Quantitative Analysis of Behavior (SQAB) in May of 2014 allowed me to reconsider some of my old papers (Marr, 1986, 1995, 2003a, 2003b) as well as new sources and provide—largely from the perspective of a behavior analyst—a more up-to-date picture of the immensely complex and surprisingly contentious practice of mathematics. The scope—breadth as well as depth—of the topic is enormous, encompassing some of the most vexing problems in our struggle to understand Nature. How has that knowledge come about and evolved, and how it is possible to grasp and predict the patterns of nature by talking about them in the special ways we call mathematics? Even more puzzling, initially many of these special ways seem merely the product of constrained verbal play, having

no apparent relations whatever to our experiences of nature. To begin to address such issues, as any behavior analyst would assert, one needs to look at the history of mathematics as well as its actual practice. With respect to practice, we might look at aspects of its acquisition and controlling variables in the light of current understanding of the dynamics of verbal behavior.

1.1. What is Mathematics?

One should not be surprised that there is no consensus on a definition of such a “motley” enterprise (cf. Wittgenstein, 1978 #46, p.176). Hacking (2014) lists 17 different definitions gleaned for various dictionaries. Davis and Hersh (1981) provide a provisional description as the *science of quantity and space including the associated symbolism*. Another definition is the *science of patterns* (Devlin, 1994). However vague, I find this latter definition appealing because the origin of all sciences is found in pattern recognition and generation, and mathematics is the most powerful method we have for characterizing those patterns we find in nature as well as generating new ones. Strongly interacting with our mathematical descriptions of nature are those patterns engendered within the exercise of mathematics itself.

Traditionally, mathematics is divided into the domains of algebra, geometry and analysis (e.g., Gowers, 2008), but this is far too simplistic, if only because of the considerable overlap of these areas in modern mathematics. Moreover, other fields not so easily clas-

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sified have emerged in the last century or so. One estimate of the number of distinguishable areas in current mathematical practice exceeds 3000, an estimate Davis and Hersh made in 1981! Clearly, no mathematician could have a practicing knowledge of anything but a very tiny percentage of such a huge edifice.

1.2. But what's it all about?

In raising this question, I am attempting to address what philosophers might describe as the *Nature* of mathematical knowledge. I raise this question because in briefly addressing it, and adopting a point of view, I think we may be led to consider the practice of mathematics in the light of a *behavioral science*—or, at the least, free ourselves from the magic and mysticism that are so often attributed to mathematics. One cannot begin to approach the analysis of mathematics as verbal behavior without a close study of how mathematicians actually work, and especially, how they have acquired their skills. Formal philosophical treatments of both science and mathematics were especially popular in much of the last century, but both efforts foundered on attempts to achieve foundations. To have any grasp of these seemingly lofty enterprises, one has to bring them down to earth through explorations of the actual practices of scientists and mathematicians—in other words, provide a *behavior analysis*. In the light of actual practices, foundations evaporate.

The philosophy of mathematics has had a long and checkered history that burned brightly with controversy in the last century, but while a good bit of that heat has dissipated, some of the issues still smolder (e.g., [Hacking, 2014](#)). Much of these concerns involve two interrelated questions: (1) what is the status of mathematical truth; and (2) what is the nature of mathematical objects? As [Gowers \(2006\)](#) expresses it, “. . . what gives mathematical statements their aura of infallibility, and what on earth are these statements about?” (p. 182). In addressing such questions, many different “schools” developed: Platonism, logicism, formalism, nominalism, intuitionism, and empiricism to name the major ones.

In the interest of brevity as well as relevance, I will focus here only on two opposing views—*Platonism* (sometimes, curiously, called realism) and *empiricism*. There are varieties of each, but I will ignore much of that detail ([Hersh, 1997](#); [Hacking, 2014](#)).

1.3. Platonism

The Platonist or realist school believes in the reality of mathematical entities as eternal, objective facts independent of human existence; in other words, something like Plato's ideal forms. While this position was only named in the 20th century ([Hacking, 2014](#)), it is of considerable age. The 19th century mathematician Hermite expressed the view as follows:

I believe that the numbers and functions of analysis. . . exist outside of us with the same character of necessity as the objects of objective reality; and we find or discover them and study them as do the physicists, chemists and zoologists (quoted by [Kline, 1980](#); p. 322).

Thus, mathematics is placed alongside other natural sciences, and the mathematician, by implication, is a *discoverer*, not a creator. As G. H. [Hardy \(1967\)](#) noted:

I believe that mathematical reality lies outside of us, that our function is to discover or *observe* it, and that the theorems which we prove, and which we describe grandiloquently as our ‘creations’ are simply our notes of our observations (pp. 123–124). [Hersh \(1997\)](#) remarks that this position is analogous to that of a botanist who can discover and classify plants, but cannot create new species.

The perspective of Platonism is compelling by any account. First, it is a view that most of us naively adopt as a result of our training

in elementary mathematics and it is probably implicit in the activities of most practicing mathematicians. As [Davis and Hersh \(1981\)](#) observe: “Platonism was and is believed by (nearly) all mathematicians. But like an underground religion, it is observed in private and rarely mentioned in public” (p. 339). We are so imbued with the reification of the integers, that it does not seem so odd to consider them as pre-existing entities, but then what about π , or $\sqrt{-1}$, or infinite-dimensional vector spaces, or Cantor's transfinite numbers, or a function discontinuous at every point? The list of such mathematical “objects” is beyond counting and ever growing! What can it mean to say that there exist mathematical entities? Clearly, one is not talking about objects in the environment (other, than perhaps, marks on paper). For the Platonist, “These objects are, of course, not physical or material. They exist outside the space and time of physical existence. They are immutable—they are not created and they will not change or disappear” ([Davis and Hersh, 1981](#); p. 318).

If we put such a view alongside Cartesian dualism, we now apparently have a *trinity*—mind, matter, and immutable, eternal forms outside of mind or matter. Actually, mind and matter would seem completely subservient to those eternal forms, for these are, in the words of one mathematician, “. . . the essence of what drives the universe” (cited in [Hersh, 1997](#), p. 10) thus, mathematics is the ultimate “intelligent design”! We may well ask as does Hersh (1997, p. 11): “Why do mathematicians believe something so unscientific, so far-fetched as an independent immaterial timeless world of mathematical truth?”

From the perspective of a competent practicing mathematician, these beliefs are perhaps not so difficult to understand as they seem inherent in the experience of doing mathematics. Certainly, one can have the strong *feeling of discovery* of one or more of these eternal entities putatively driven by what [Wittgenstein \(1978\)](#) called “the hardness of the logical must.” What is not being brought to bear on these beliefs is *history*—both the history of mathematics over millennia, and, at least as important, the individual's history in acquiring and practicing their mathematical craft. I will return to these elements shortly. But now for an alternative perspective of the content and practice of mathematics—*empiricism*.

1.4. Empiricism

As opposed to believing that the universe is somehow built from pre-existing mathematical forms and relations which we discover, one can turn this perspective on its head by proposing that nature evinces patterns in its phenomena and that we *invent* the means to describe those patterns. Why does mathematics work so well? This issue was addressed in a classic essay by [Wigner \(1984\)](#) entitled “The unreasonable effectiveness of mathematics in the natural sciences”. So, what is the problem here? Let me start with an anecdote from Wigner. A statistician was explaining to a non-mathematical friend the Gaussian, or normal distribution describing characteristics of a population:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The friend was skeptical and asked “How can you know that?” “And what is this symbol here?” “Oh,” said the statistician, this is π .” “What is that?” “The ratio of the circumference of a circle to its diameter.” “Well, now you are pushing the joke too far—surely the population has nothing to do with the circumference of a circle” (p. 117). Not only is it the case that mathematical concepts turn up in unexpected places, but physicists, engineers, and others are able to develop mathematical models of startling effectiveness both in describing nature's contingencies and predicting previously unobserved phenomena. To Wigner and to many others who have practiced this art, it “borders on the mysterious and there. . . is no

COMPLEX VARIABLE EVOLUTION

1. Quadratic/cubic equations: $x^2 + 1 = 0$
2. Wessel's complex plane: $a + bi = \sqrt{a^2 + b^2} \angle \tan^{-1} \left(\frac{a}{b} \right)$
3. De Moivre's theorem: $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$
4. Euler's identity: $e^{i\theta} = \cos \theta + i \sin \theta$
5. Riemann's zeta function: $\zeta(z) = \prod_p \left(1 - \frac{1}{p^z} \right)^{-1} = \sum_1^\infty \frac{1}{n^z}, \text{Re}(z) > 1$
6. Contour integration: $\oint_C f(z) dz = 0$
7. Conformal mapping: $w = z + \frac{1}{z}$
8. Schrödinger equation $H(\Psi) = i \frac{h}{2\pi} \frac{\partial \Psi}{\partial t}$
9. Mandelbrot set: $M : P_c(z) = z^2 + c$

Fig. 1. Developments in complex variable theory over centuries starting with (1) efforts to solve quadratic and cubic equations involving complex roots; (2) representing complex numbers in the plane; (3) and (4) showing relations between complex numbers and trigonometric and exponential functions; (5) Riemann's studies of the distribution of prime numbers; (6) using complex variables to solve difficult definite integrals; (7) mapping or transforming regions from one complex plane to another, for example, to achieve a simpler representation for analysis; (8) quantum mechanics uses many functions involving complex expressions; (9) iterated function systems with complex variables can produce elaborate fractal structures like the Mandelbrot set.

rational explanation for it" (p. 117). Wigner concludes his essay with the mystery remaining: "The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve" (p. 124).

How indeed, do we come to know so much about Nature by talking about it? The answer seems mind-bogglingly simple: *Because nature taught us how to talk*. Emo Phillips, a comedian known now mostly to us crossword puzzle solvers, once commented that he used to think that the most wonderful organ in the universe was the human brain—then he realized what was telling him that! How is it that natural selection yielded up fish with streamlined bodies, birds whose wings are structured in accordance with Bernoulli's Principle, and elephants with log-like legs? Whatever the origins of verbal behavior, the outcome is a class of behavior in accordance with and capable of describing the Nature that shaped it. As behavior analysts, we characterize verbal behavior as that whose effectiveness is mediated through other people—the verbal community. But this verbal community lives in the world, the world independent of our representations of it; of course, this in no way implies that our representations are independent of the world. Hersh (1997) succinctly characterizes this view as follows:

Mathematics is part of human culture and history, which are rooted in our biological nature and our physical and biological surroundings. Our mathematical ideas in general match our world for the same reason our lungs match earth's atmosphere (p. 17).

1.5. Shaping through history

The empirical view requires a more careful justification, especially in confronting such a compelling and venerable stance as Platonism. Here I appeal to what I call the *shaping of abstraction*

and what Phillip Kitcher (1983) (p. 92) argued as an *evolutionary theory* of mathematical knowledge. But with any appeal to shaping one must investigate how the final performance has been acquired.

The history of mathematics reveals what one would expect, namely that the rudiments emerged through socially significant needs—bartering, land and other property measuring, building, farming, sailing, conducting rituals, and numerous other practices. The literature here is huge, and there are numerous examples, each worthy of a large volume (e.g., Kline, 1972; Katz, 2009; Stillwell, 2010). Perhaps the most familiar case takes us over millennia from simple counting to what might be called *number systems*—natural numbers (that is, the positive integers), fractions (the rationals), the invention of 0, negative numbers, irrational numbers, complex numbers, quaternions, transfinite numbers, and so on—each a *new concept*. In the context of number systems, I will choose a single example for a superficial survey—complex variables (see Fig. 1).

Complex variable theory is often acknowledged as one the most beautiful branches of mathematics (e.g., Ablowitz and Athanassios, 1997; Nahin, 1998). Modern applications include fluid dynamics, electrical engineering, heat flow, relativity, quantum mechanics, cartography, iterated function systems, number theory itself, and many others. The considerable (and challenging) mathematical developments in this theory have a deep history that can be traced back to very ancient times to concerns regarding area and volume determinations—obviously of considerable commercial interest. Initially these must have been estimated through various rules of thumb, but through persistent and systematic efforts, the details of which are entombed in history, problems of this sort began to take on a life of their own. We find, for example, in ancient tablets and papyri explicit practical problems presented for solution, the methods of which much later were to become systemized as alge-

bra and geometry each with more extended applications, but just as importantly, became areas of study in themselves. For example, attempts to find the volume of a pyramid led to the problem of the volume of a truncated pyramid which, over considerable time, led to wrestling with quadratic and cubic equations. Development of methods to solve them led in turn over many centuries to the study of complex numbers and their properties, including their geometric representation, their role in greatly extending concepts of known (and very useful) functions such as sinusoids and exponentials and their interrelations, and, along the way, in solving very difficult problems in the applications I have already mentioned.

In the process of developing these areas and many, many others, much of the work depended on purely *verbal consequences* without any clear concerns with applications. In a similar way, writers whether of poetry, drama, short stories, novels, or expository compositions such as you are currently reading, are sensitive to the consequences of verbal play—each with its own constraints. The history of mathematics compellingly reveals a continual interplay between problems in application (e.g., in engineering) and what might be called mathematical play, the latter often driven by intense competition among the players. I do not think it is difficult to see that such a dynamic enterprise with multiple feed-back loops has a snow-ball effect accumulating more and more concepts, methods and applications through innumerable variations and their consequences.

The patterns driving the evolution of concepts and practice involve a variety of verbal functions including idealization, simplification, generalization, unification, analogy, systemization, application, reconstruction, and developments in notation (for a recent history of mathematical notation, see Mazur, 2014).

Fig. 2 illustrates the specific example of the generalization of the concept of length or distance.

In this discussion of shaping through history, including individual history, you might keep in mind Arthur C. Clarke's maxim that to us the activities and products of any sufficiently advanced civilization would be indistinguishable from magic. In the case under discussion, the "unreasonable effectiveness of mathematics" would not only seem like magic, say, to Archimedes, but, for many today still "borders on the mysterious."

2. Skinner's "empirical epistemology"

That scientific behavior is principally verbal is, of course, a truism. Skinner (1957, 1969) seems to have been a pioneer in the analysis of scientific verbal behavior. He sets the stage for his discussion with the following:

When a speaker accurately reports, identifies, or describes a given state of affairs, he increases the likelihood that a listener will act successfully with respect to it, and when the listener looks to the speaker for an extension of his own sensory capacities, or for contact with distant events, or for an accurate characterization of a puzzling situation, the speaker's behavior is most useful if the environmental control has not been disturbed by other variables (1957, p. 418).

Later, he focuses down on the scientific community itself:

The scientific community encourages the precise stimulus control under which an object or property of an object is identified or characterized in such a way that practical action will be most effective (1957, p. 419).

He finishes his chapter with a remarkable conclusion:

One of the ultimate accomplishments of a science of verbal behavior may be an empirical logic, or a descriptive and analytic scientific epistemology, the terms and practices of which will be adapted to human behavior as a subject matter" (1957, p. 431).

Skinner treats various classes of verbal operants common to scientific discourse and practice, the manipulation of verbal material, the role of graphs, models, mathematical, and logical operations. Subsequently (1969; 1974), he discussed the roles of what he termed rule-governed and contingency-controlled behavior, which I will address shortly.

2.1. The role of verbal operants

The modes of verbal evolution in the long history of mathematics previously mentioned as well as in individual mathematical practice find at least some representation in Skinner's account, in particular as functioning categories of verbal operants and their controlling variables. With respect to what Skinner calls a *tact*, there are generic and metaphorical extensions, as well as abstraction. I

GENERALIZATION OF "DISTANCE"

right triangle: $s^2 = a^2 + b^2$

any triangle: $s^2 = a^2 + b^2 - 2ab \cos \theta$

n -dimensional Euclidian space: $s^2 = \sum_{i=1}^n x_i^2$

Length of a curve: $s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

Riemannian space: $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$

Fig. 2. The quantitative developments in the concept of length or distance from the Pythagorean theorem in two dimensions to an n -dimensional Euclidian space. This was the foundation for further generalizations. With the development of the calculus, lengths of any well-behaved curves could be calculated. Finally, distance in any space could be determined by knowing the "little g 's"—the metric tensor characterizing the space. This is a gateway to Einstein's General Relativity Theory.

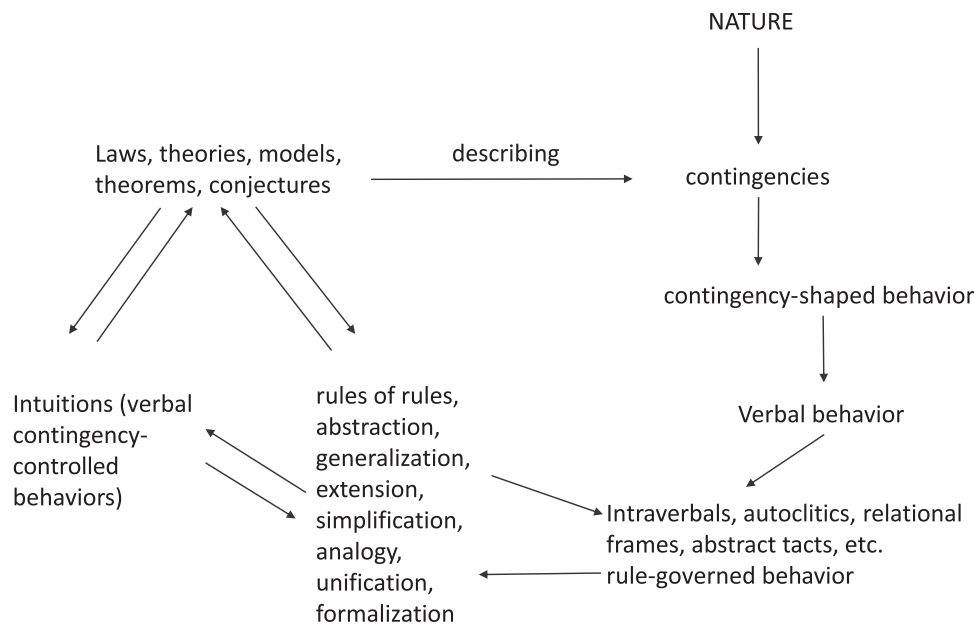


Fig. 3. A scheme summarizing the main points of this paper. See Section 3 for details.

remind the reader that a tact is defined as: "... a verbal operant in which a response of a given form is evoked (or at least strengthened) by a particular object or event or property of an object or event" (1957, pp. 81–82). Thus, basically the tact has the function of a discriminative stimulus in a three-term contingency and, most significantly, displaces the traditional roles of "meaning," or "reference." But especially given the problems of Platonism, tacts should have a limited, though perhaps necessary, role in mathematical verbal behavior. A more important operant is the *intraverbal*, that is, a verbal operant occasioned by a current or previous verbal operant: $2 + 2 = 4$ is a simple example. Skinner recognized its significant role when he asserted: "A great deal of scientific, mathematical, and logical discourse is... intraverbal and hence not adequately represented by the semantics of the tact" (1957, p. 129). An even greater function is served by various forms called *autoclitics*, which are, in a certain sense, extensions of intraverbals in that the controlling relations are occasioned by verbal behavior itself—its description, modification, ordering, qualification, and quantification—much of what traditionalists might call "grammar." Quoting Skinner [On autoclitics] "... assertion, negation, and quantification, the design achieved through reviewing and rejection or emitting responses, the generation of quantities of verbal behavior merely as such, and the highly complex manipulations of verbal thinking can all... be analyzed in terms of behavior which is evoked by or acts upon other behavior of the speaker" (1957, p. 313). Palmer (2013) in recent paper on verbal behavior in logic and mathematics places proper emphasis on *autoclitic frames*; to quote his definition, "... an autoclitic frame is a interrupted or incomplete intraverbal sequence into which variable terms are inserted according to context" (Palmer, p. 273). Any mathematical text will illustrate hundreds of examples: *Given X, show that Y; If and only if X, then Y; Find X, given Y; X is greater than or equal to Y*, and countless others are an essential part of the day-to-day verbal repertoire of the mathematician.

2.2. Contingencies, rules, and relations

At least some of the above expressions can be formulated as rules; and rules, indeed, rules of rules, play a fundamental part of the verbal behavior of the mathematician. To quote Wittgenstein, "[Mathematics] forms ever new rules; it's always building new roads for traffic; by extending the network of the old ones"

(RFM, #166, p.99). But the term "rule" is a slippery one and over the years many behavior analysts have wrestled with the concept. Skinner described rules as verbal discriminative stimuli specifying contingencies, and contrasted behavior under control of such stimuli, that is *rule-governed behavior*, with *contingency-controlled behavior*. In the latter case, behavior is said to be under *direct* control of its consequences. As an alternative to the possible discriminative control said to adhere to a rule, Schlinger argues that a rule may instead have a function-altering effect; and, considering the many ways the word is used, he ultimately argues for dispensing with the term "rule" altogether (Schlinger, 1993, 2013). Aside from that position, Dan Cerutti in a brilliant 1989 review discussed rules in the context of instructional control and proposed that instructions can engender generalized discriminative and response classes. Here is a quote relevant to my topic:

Complex responses... may become generalized response classes as in the instructed problem-solving strategies in geometry, calculus, and chemistry... Such instructed problem-solving units are also generalized classes, to the extent they retain their integrity as the transfer to problems with novel combinations of elementary features (Cerutti, 1989; p. 263).

The sorts of extensions I discussed earlier can easily fit into a scheme such as Cerutti describes, and these sorts of verbal dynamics are evident, for example, by a mathematically competent student solving a complex problem, say in electromagnetics. What one sees are deeply nested and hierarchical repertoires of methods, concepts, rules, strategies, intuitions, all these and more skills brought to bear on the given problem and all the outcome of an extended history. This leads me to my final topic—intuition in mathematics.

3. Intuition

"... behaving intuitively in the sense of behaving as the effect of unanalyzed contingencies, is the very starting point of a behavior analysis" (Skinner, 1974; p. 132). We do have some idea of the conditions necessary for the development of intuition in the expression of complex skills of virtually any kind. The biographies of eminent mathematicians and scientists all reflect an essential common theme—persistent, intense, obsessive, disciplined, and devoted practice of their craft. For example, Richard Feynman

as a young high-school student learned mathematics by working through standard texts in trigonometry and the calculus, keeping elaborate notebooks of carefully worked problems. Rather than simply looking up or copying table values and functions, he calculated logarithmic and trigonometric values directly and derived tables of integrals for himself (Mehra, 1996).

Srinivasa Ramanujan, arguably the 20th century's greatest mathematician (and certainly the most mysterious), was already on his way at age 16 having spent virtually all his waking hours doing mathematics and with very little tutoring. Here is how his biographer Kanigel (1991) described Ramanujan's methods in number theory:

Even in his published notebooks, you can see Ramanujan giving concrete numerical form to what others might have left abstract—plugging in numbers, getting the feel for how functions “behaved”... Numerical elbow grease it was... Ramanujan was doing what great artists always do—diving into his material. He was building an intimacy with numbers, for the same reason that the painter lingers over the mixing of his paints, or the musician endlessly practices his scales (p. 63).

What are the consequences of such persistent, relentless, compulsive play? Simply stated, a huge behavioral repertoire with complex and flexible units, intricate associative links, autoclitic and relational frames, all in a dynamic blend. No wonder persons with these histories can seem magical in their powers. But it is a magic gestated in the womb of toil, untold thousands of hours of it. Even the highly skilled individual may not appreciate how and what such a history may engender. For example, the literature on creativity is replete with autobiographical descriptions of creative acts that seem startling to the creator, as well as others, because the sources are lost to them in a sea of experience. From this sea emerges *intuition*.

Such training yields “automaticity” in application of rules, concepts, and other verbal relations in the generation and application of mathematics—what the relational frame people might call emergent mutual entailment, combinatorial entailment, and transfer of function (Hayes et al., 2001). Through such training, what begin as rules, for example, may be transmuted into contingencies having direct consequences on behavior that, in turn, can yield new rules and concepts. When rules and associated concepts come to act as contingencies, we call the outcomes “intuition.” According to this view, intuition is an expression of a kind of contingency-controlled behavior—in the case of mathematics, verbal contingencies, with not only verbal (and perhaps even non-verbal) consequences; but, essential and more elusive forms of reinforcement one can only describe as aesthetic. Achieving control over verbal material can be at least as powerful as achieving control over any non-verbal domain, and mathematics has the advantages of doing both. A layperson may imagine the great mathematician at work starting with a set of rules, and step-by-step, generating one term or theorem after another in a kind of intraverbal chain of logic leading toward some final proof. This is utterly mistaken as the Fefferman attests:

The mathematician at work relies on surprisingly vague intuitions and proceeds by fits and starts with all too frequent reversals. Clearly logic as it stands fails to give a direct account of either the historical growth of mathematics or the day-to-day experience of its practitioners (cited in Davison and Hersh, 1981 p. 357).

4. Coda

Fig. 3 is a digest of what I have discussed in this paper. Nature's contingencies have led to the uniquely complex human behavior

we call verbal and thus to descriptions of those contingencies in modes we call rules as well as a host of other verbal functions, all of which have enormous consequences for the development of civilizations. Such repertoires and their consequences extend those accomplishments into the development of special verbal repertoires we call mathematics. The concepts and methods of this field have evolved through selective consequences, some purely verbal, a process I call the shaping of abstraction. In many ways, these processes are repeated in the development of knowledge and skills in the individual mathematician. With extended experience in this domain, behaviors with the character of contingency control emerge—what is called mathematical intuition. Finally, sufficiently developed mathematical repertoires provide further descriptions of Nature's contingencies in a kind of cycle—building up the snowball.

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